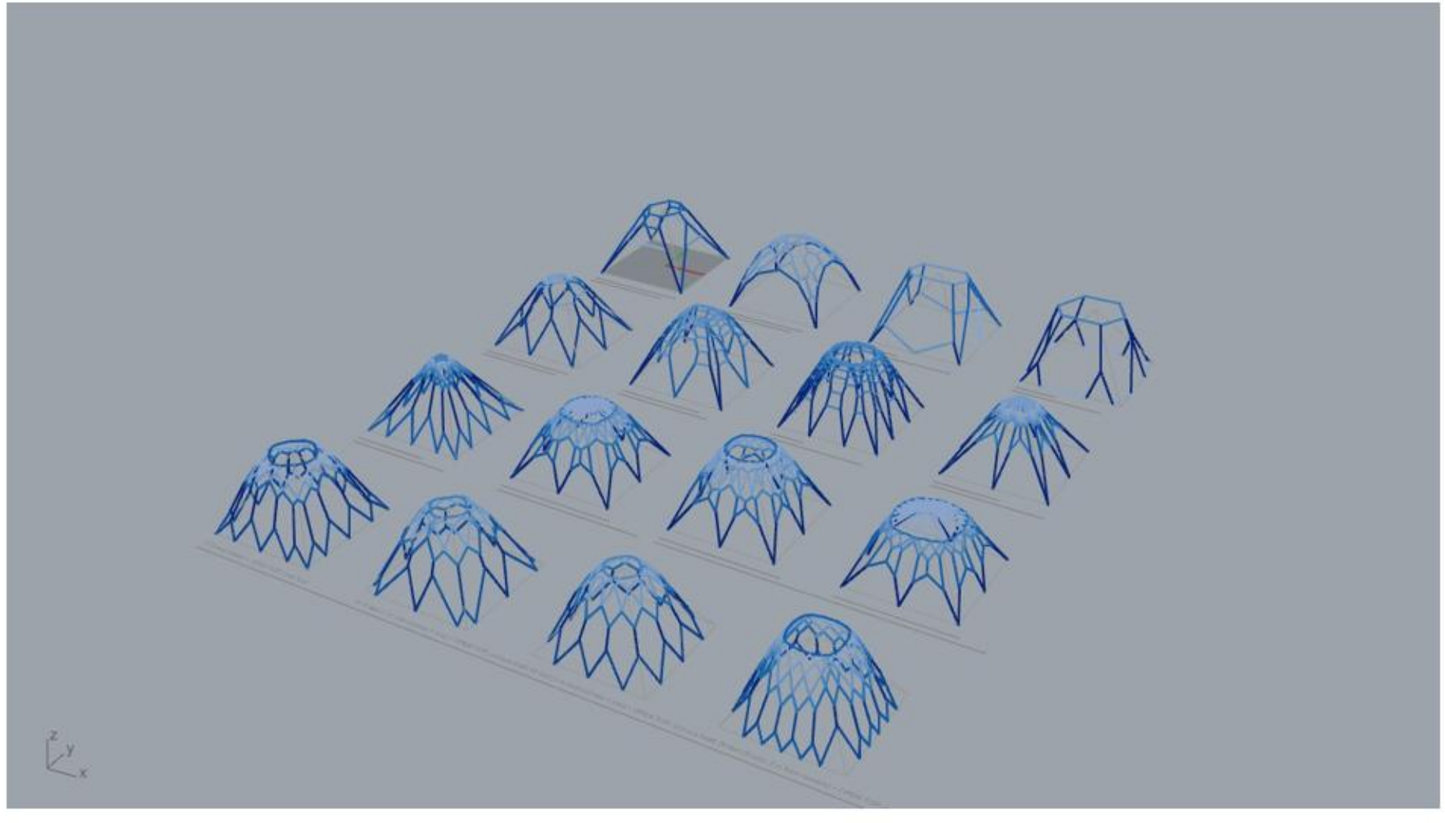
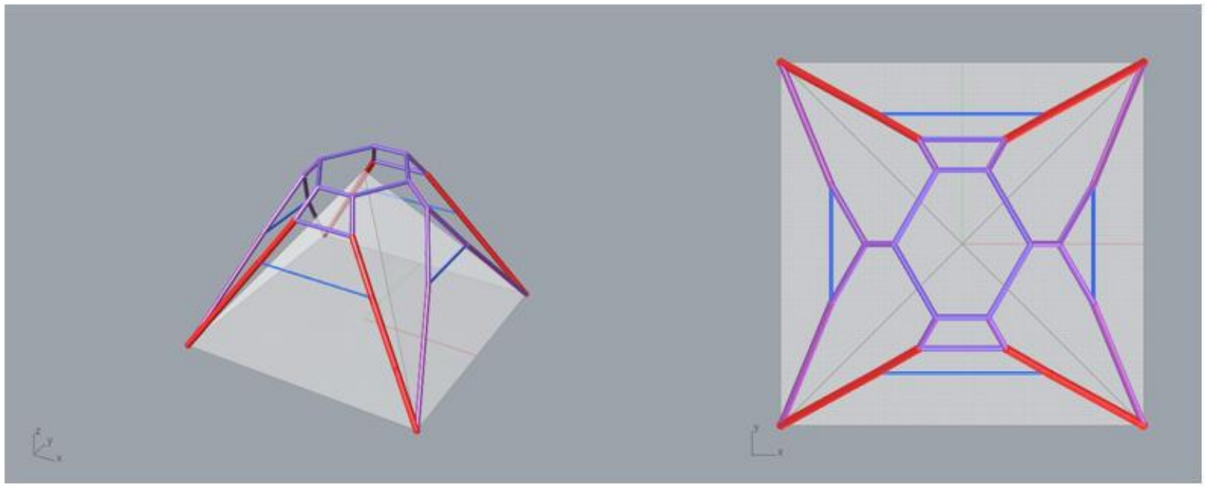
Friend's roof variants - all iterations (PolyFrame 2, 10x10 m plan, apex ~6.67 m)

All files in polyframe2 (not committed). 3D review: friend-iterations-review-v5.3dm, one layer per iteration. C = compression.
Karamba: cast C30/37, ribs 150x250, rigid joints, feet pinned, self-weight + 7 kN/m2 (assumed) on the pyramid surface; elastic stress check, no rebar.

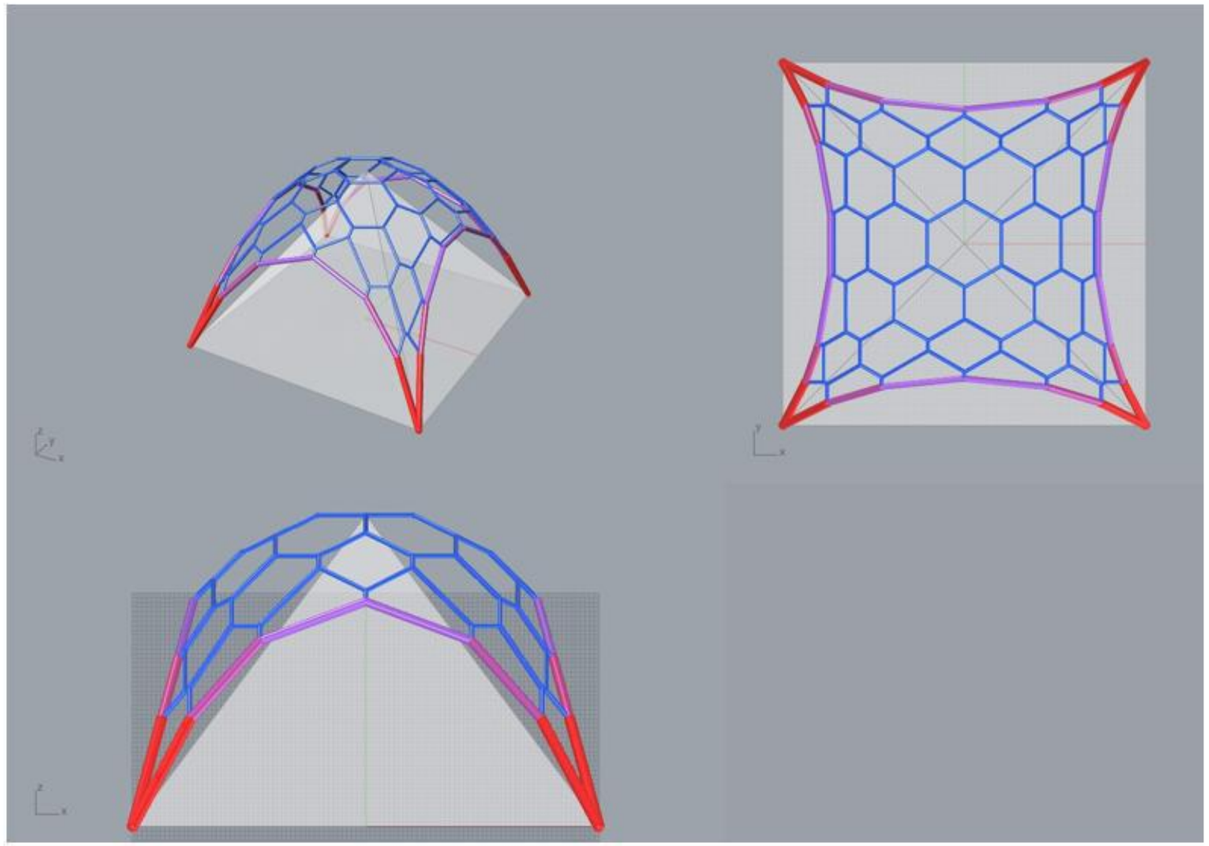
#	Iteration	Feet	PolyFrame final shape	Karamba comp/fcd	Bars <=35 deg	Known flaw	File
1	Panel A frame	4	exact, all C	-	-	not the sheet's look	friend-A-4corners
2	Panel A shell (ex 3(7))	4	exact, all C (136 bars)	-	-	free edges 1.3 m vs 2.3 m	friend-A-shell
3	AP8 frame	8	stage B all C	-	-	not the sheet's look	friend-AP8-buttresses
4	AP12 frame	12	stage B all C	-	-	not the sheet's look	friend-AP12-buttresses
5	AP8 shell GUESS	8	stage B all C	-	-	not symmetric	friend-AP8-shell-GUESS
6	AP8 shell v2 (rings)	8	stage B all C	-	24/248	96 horizontal ring bars	friend-AP8-shell-v2 (+printcast)
7	AP12 shell	12	stage B all C	-	112/516	rings horizontal; stage A tension	friend-AP12-shell
8	AP8 diagrid	8	stage B all C	1.03 (corner legs)	24/272	stage A 40 tension	friend-AP8-diagrid
9	16-feet diagrid (5/side)	16	A and B all C	1.17 (mid-face)	16/192	mid-side feet idle	friend-16-diagrid
10	AP8 even load (hub)	8	A and B all C	0.58	24/272	48 flat hub bars; ring overlap	friend-AP8-diagrid-evenload
11	AP8 even load OPEN TOP	8	stage B all C	0.52	32/296	nodes 20 mm apart at ring	friend-AP8-diagrid-evenload-opentop
12	AP8 even load 4 rings	8	stage B all C	0.90 (corner legs)	40/224	32 flat bars; stage A tension	friend-AP8-diagrid-evenload-4ring
13	16 FEET (5/side) + OPEN TOP	16	A and B all C	0.52	36/176	4 tiny 14 mm bars at ring	friend-16-diagrid-evenload-opentop
14	12 FEET near corners + OPEN	12	A and B all C	0.56	20/132	corners 0.73 m short; net 2.6 m above pyramid	friend-12-feet-near-corners-opentop
15	**12 FEET 2 m from corners + OPEN**	**12**	**A and B all C**	**0.53**	**20/132**	**corners 1.43 m short**	**friend-12-feet-2m-corners-opentop**
16	**20 FEET 2 m from corners + OPEN**	**20**	**B all C (A tension)**	**0.37**	**76/260**	**corners 1.43 m short; nodes 54 mm**	**friend-20-feet-2m-corners-opentop**



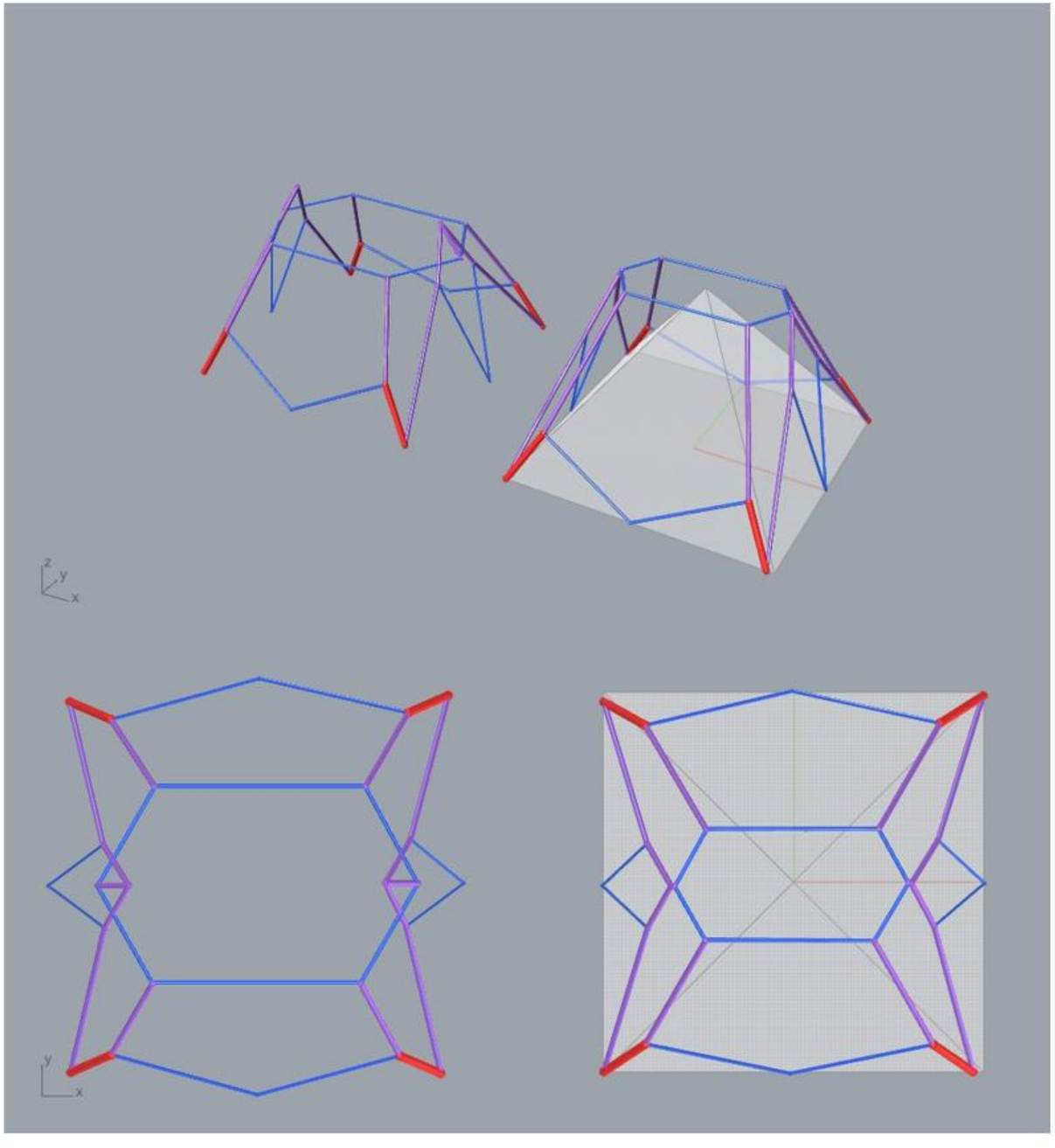
1 Panel A frame



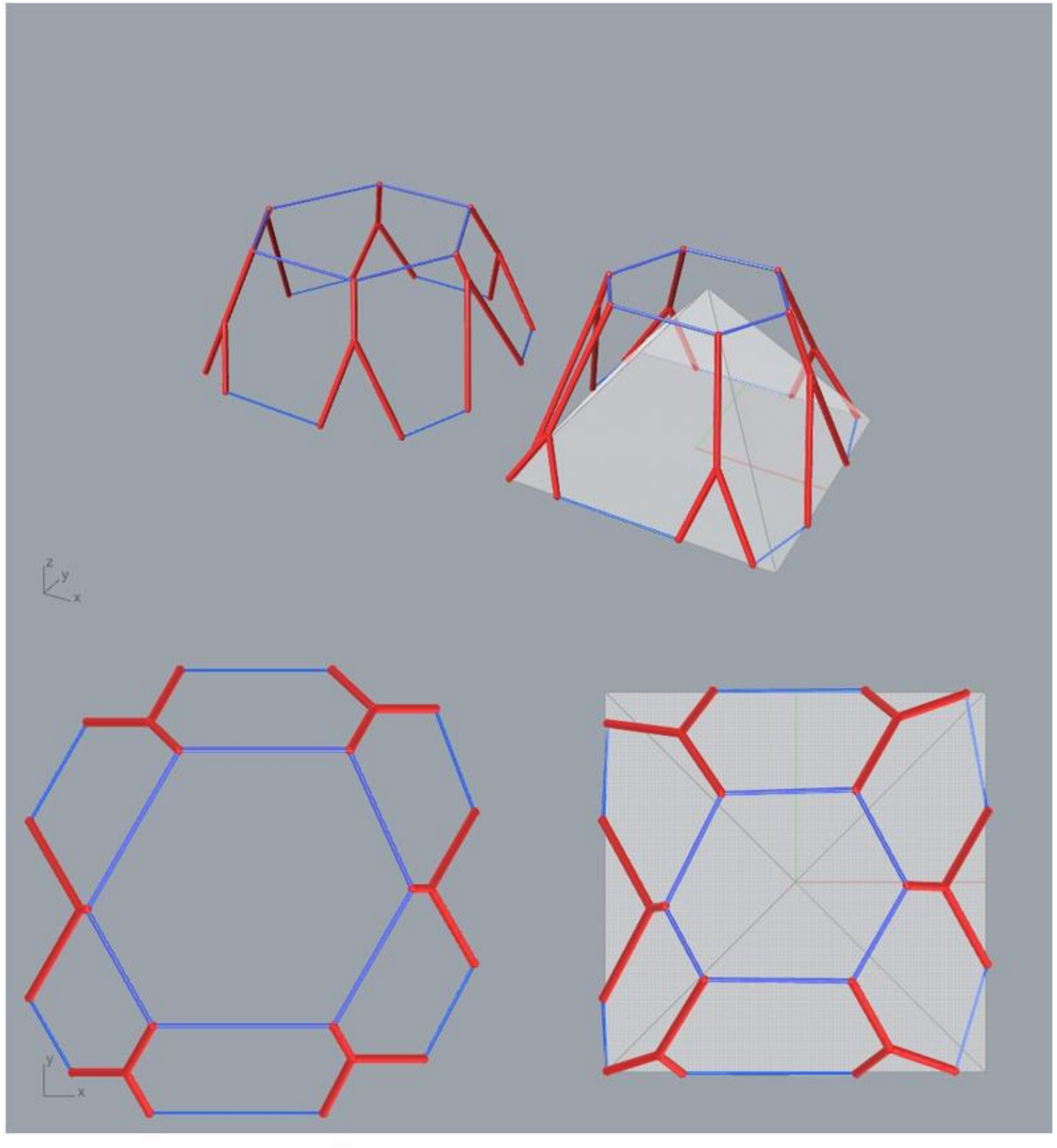
2 Panel A shell (PolyFrame example 3(7))



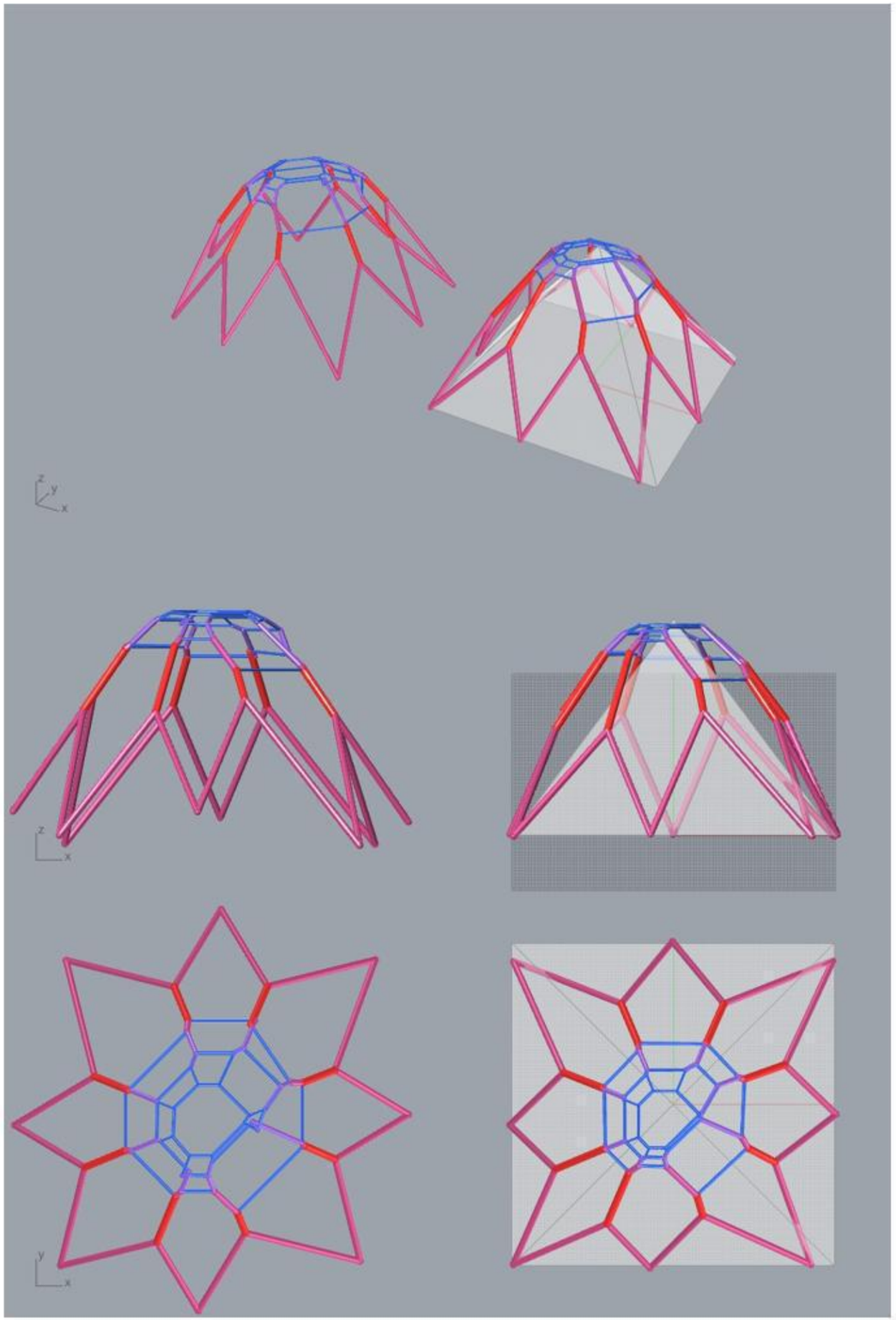
3 AP8 frame



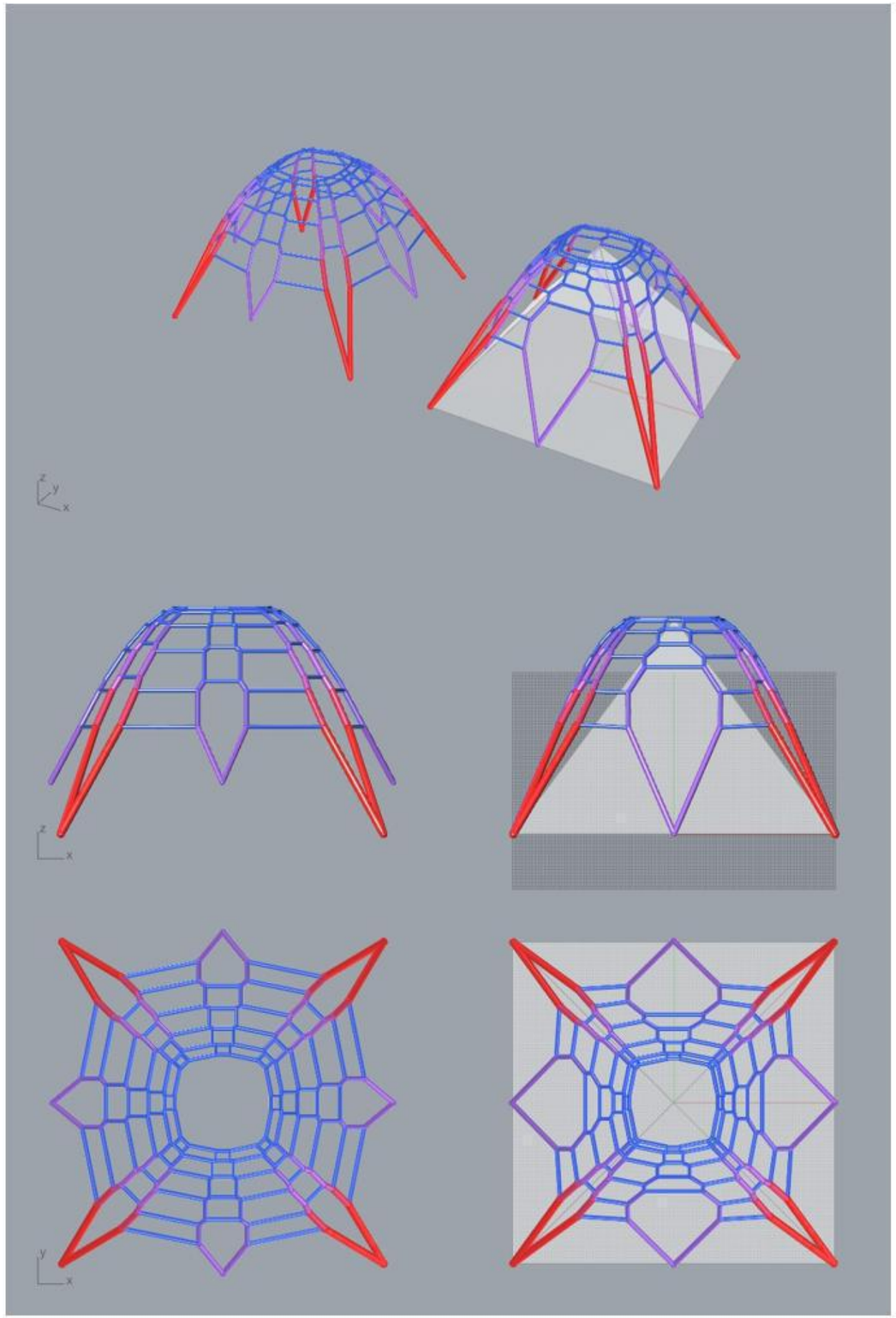
4 AP12 frame



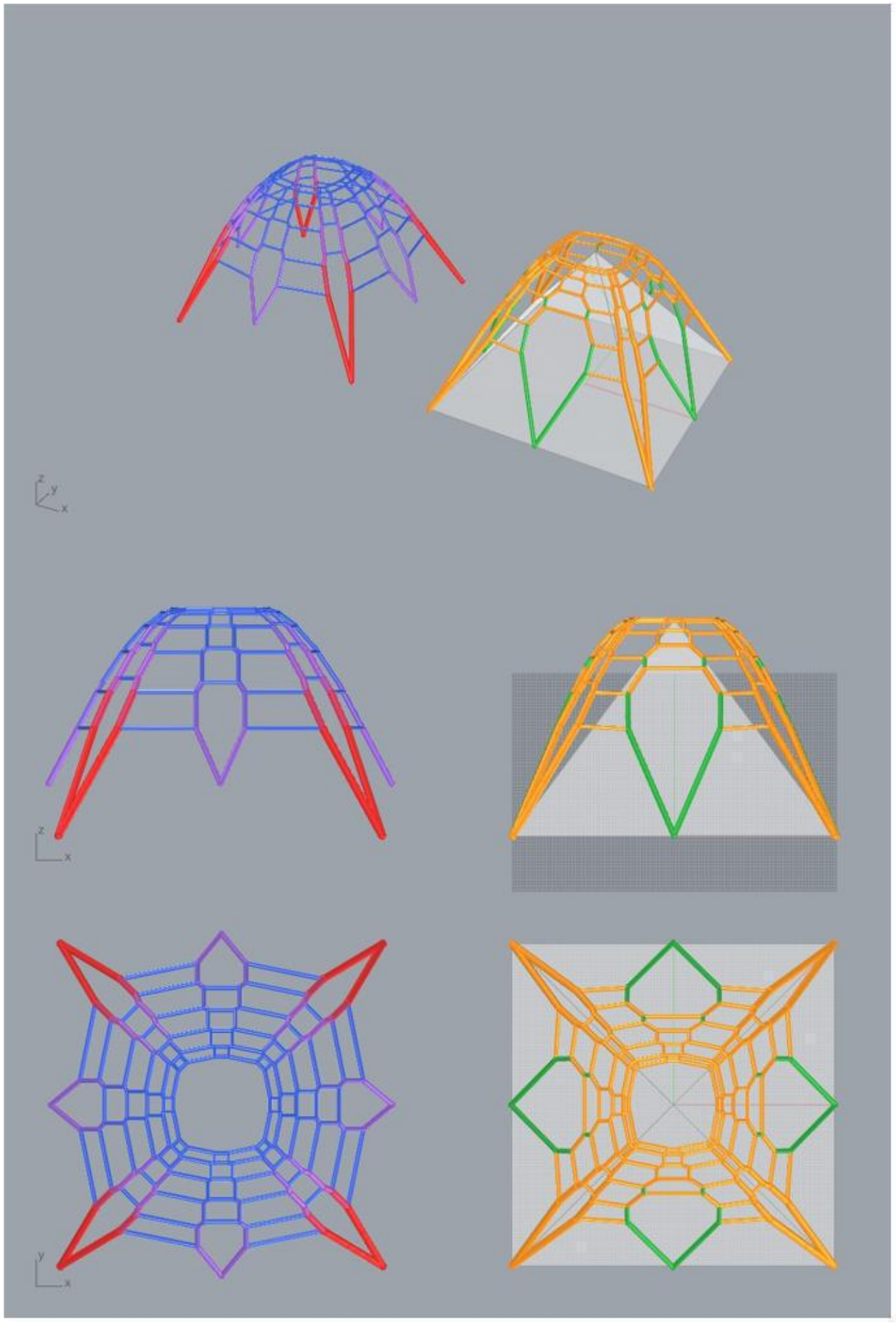
5 AP8 shell GUESS



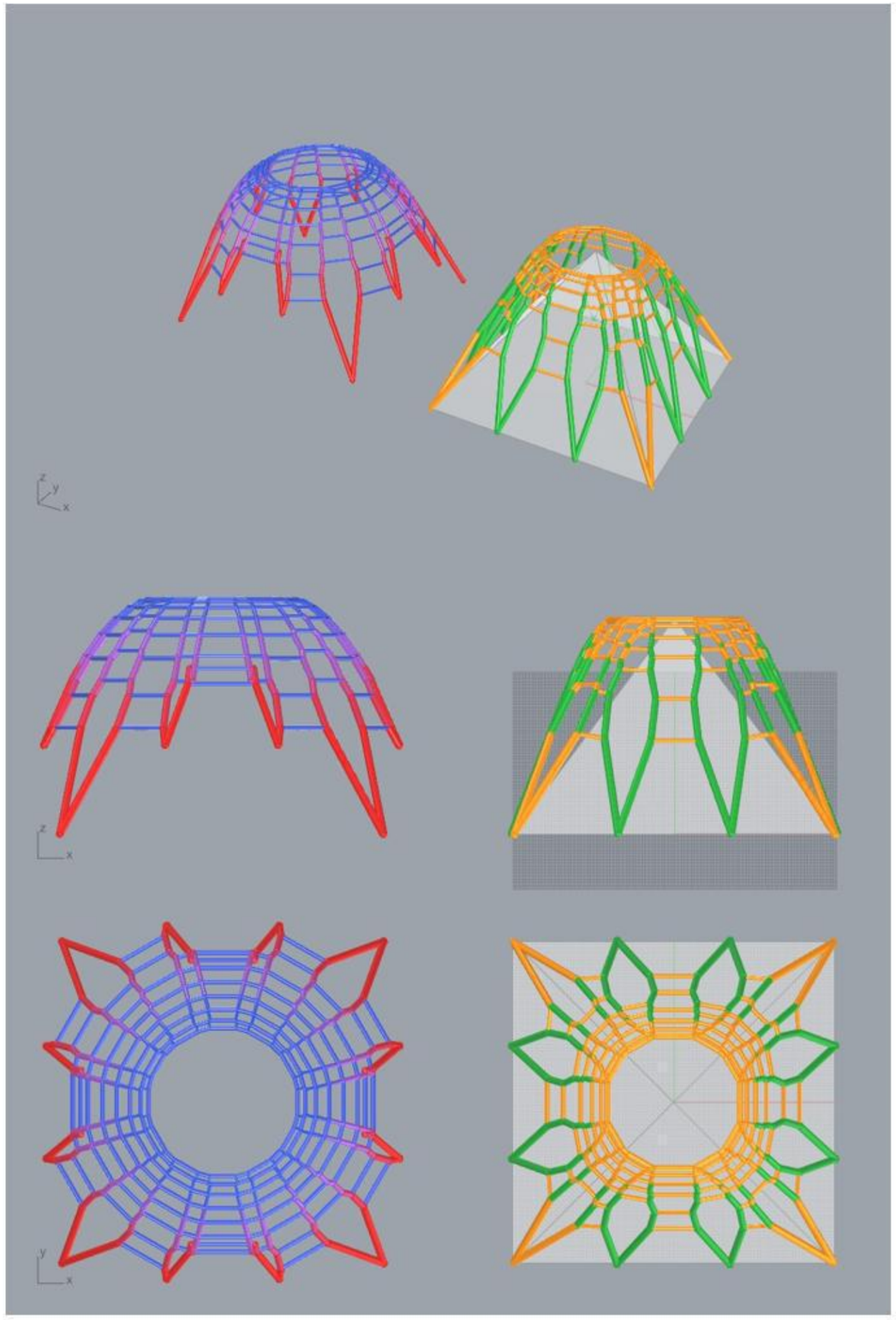
6 AP8 shell v2



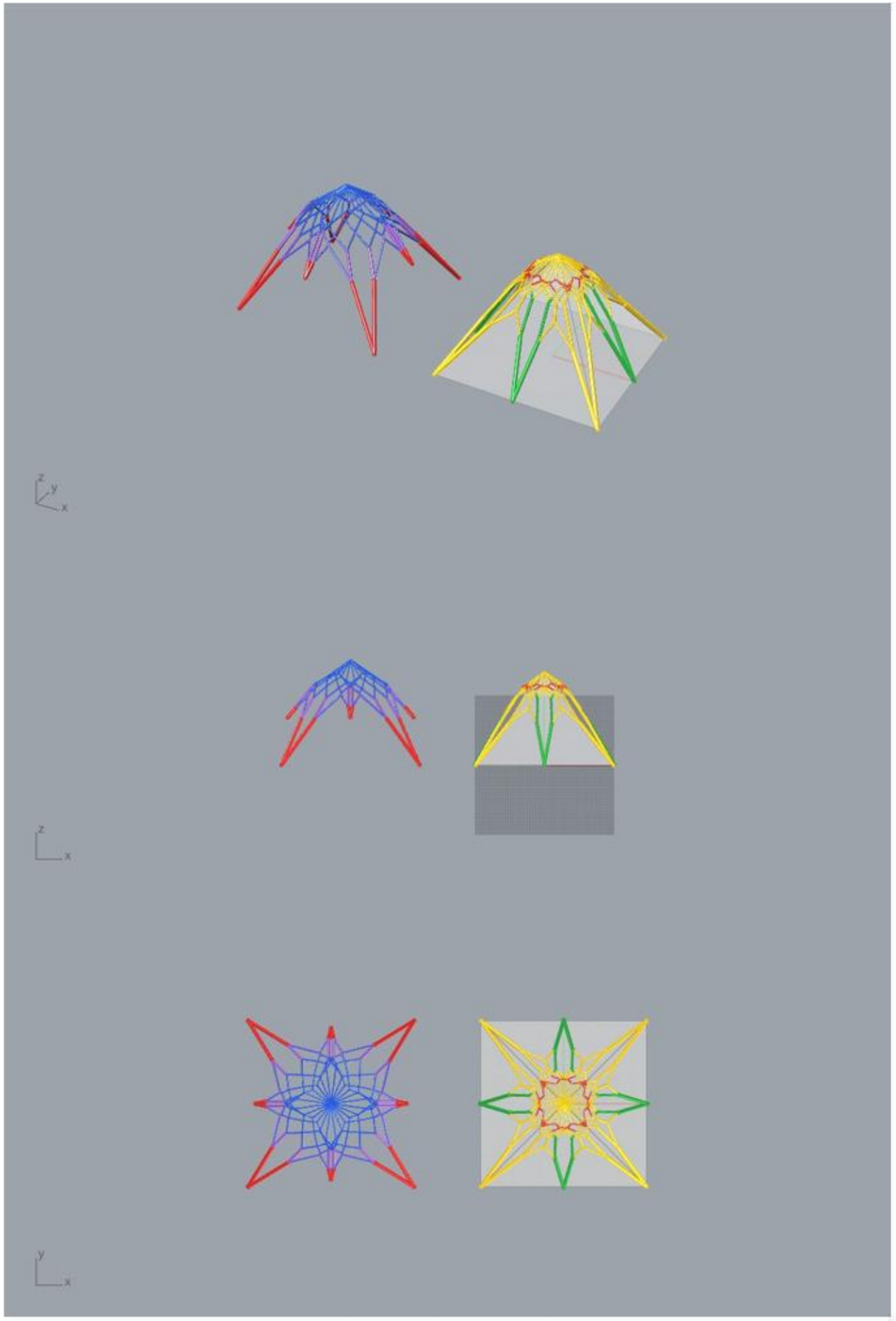
6b AP8 shell v2 - print (green) / cast (orange)



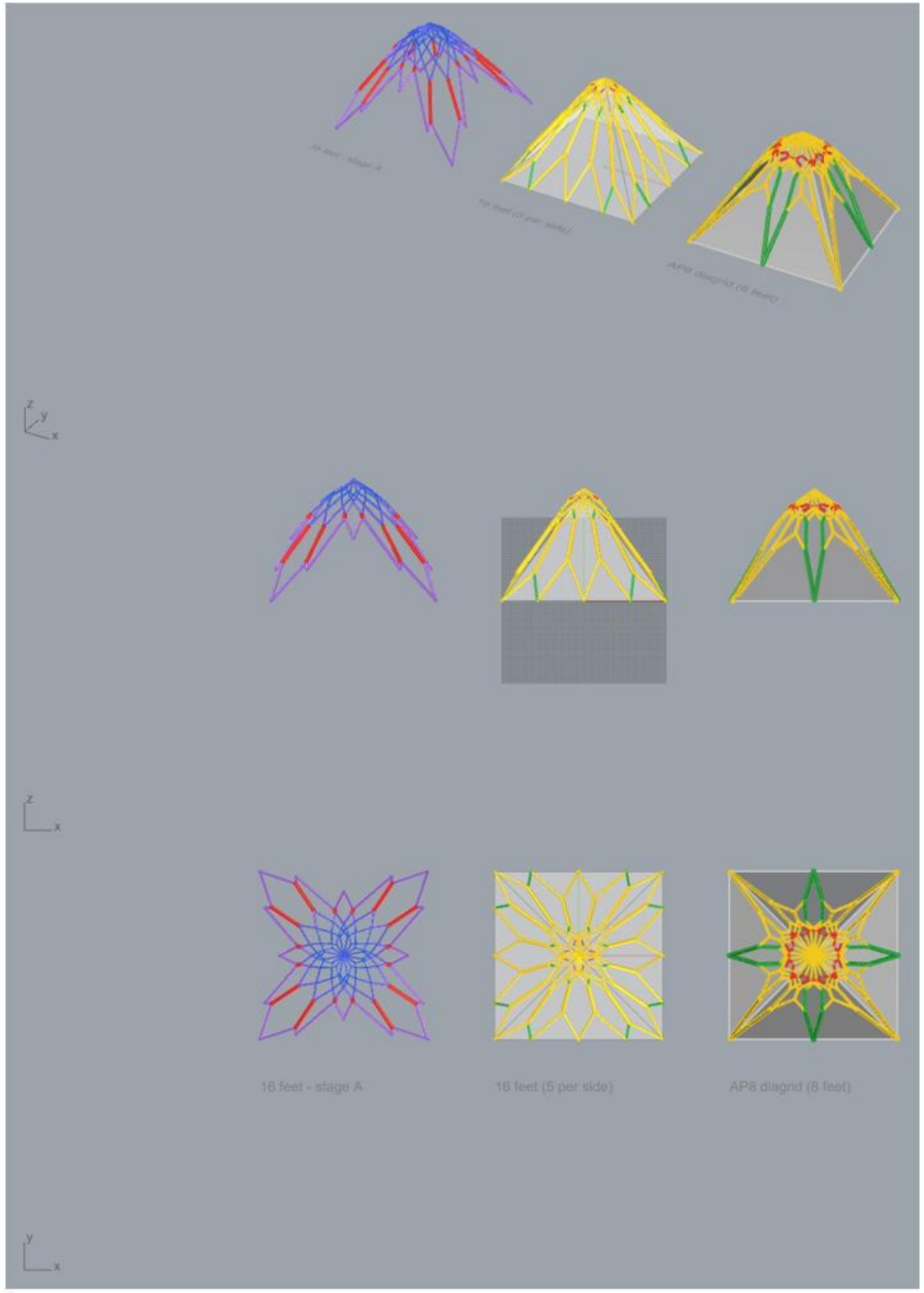
7 AP12 shell



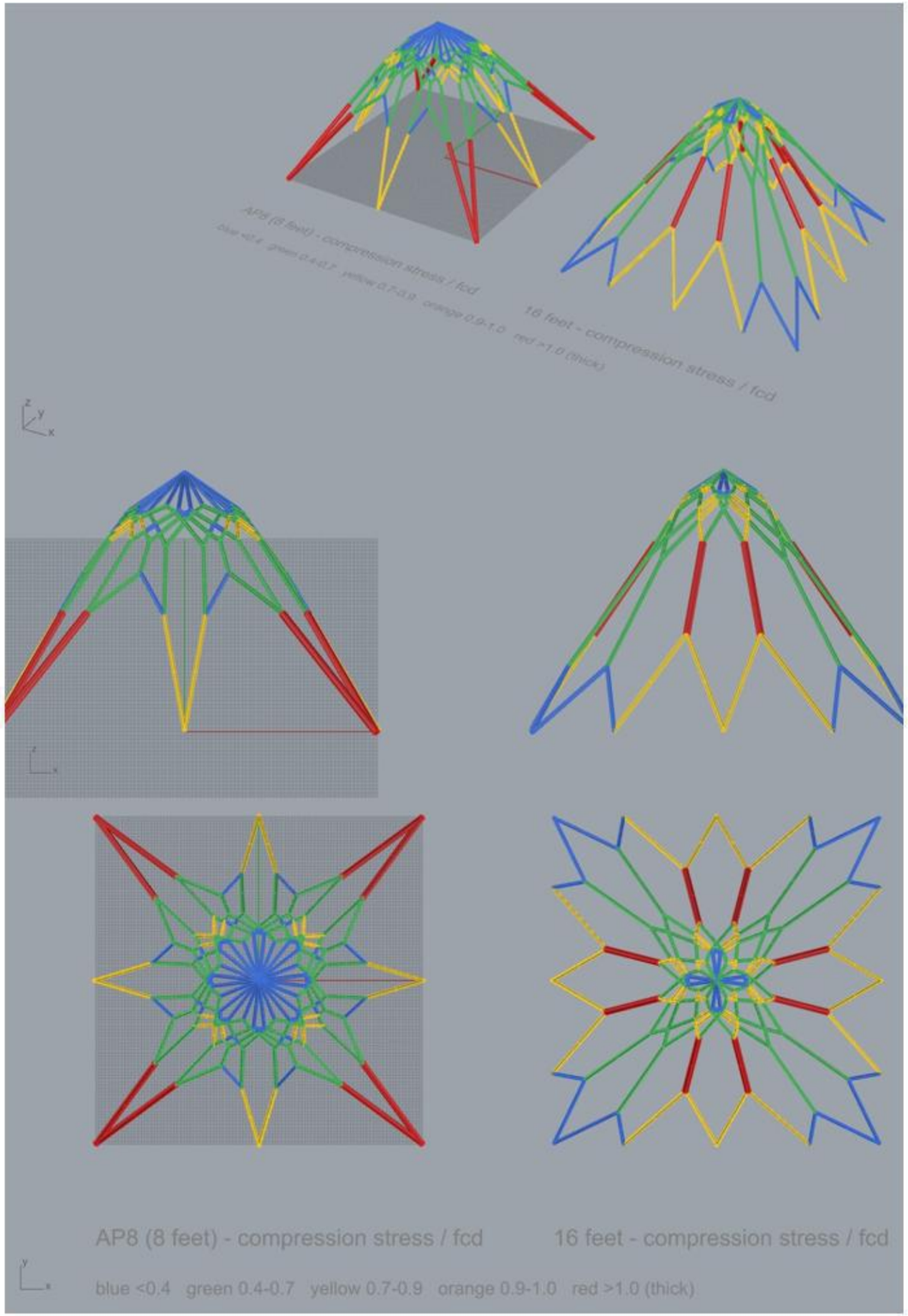
8 AP8 diagrid



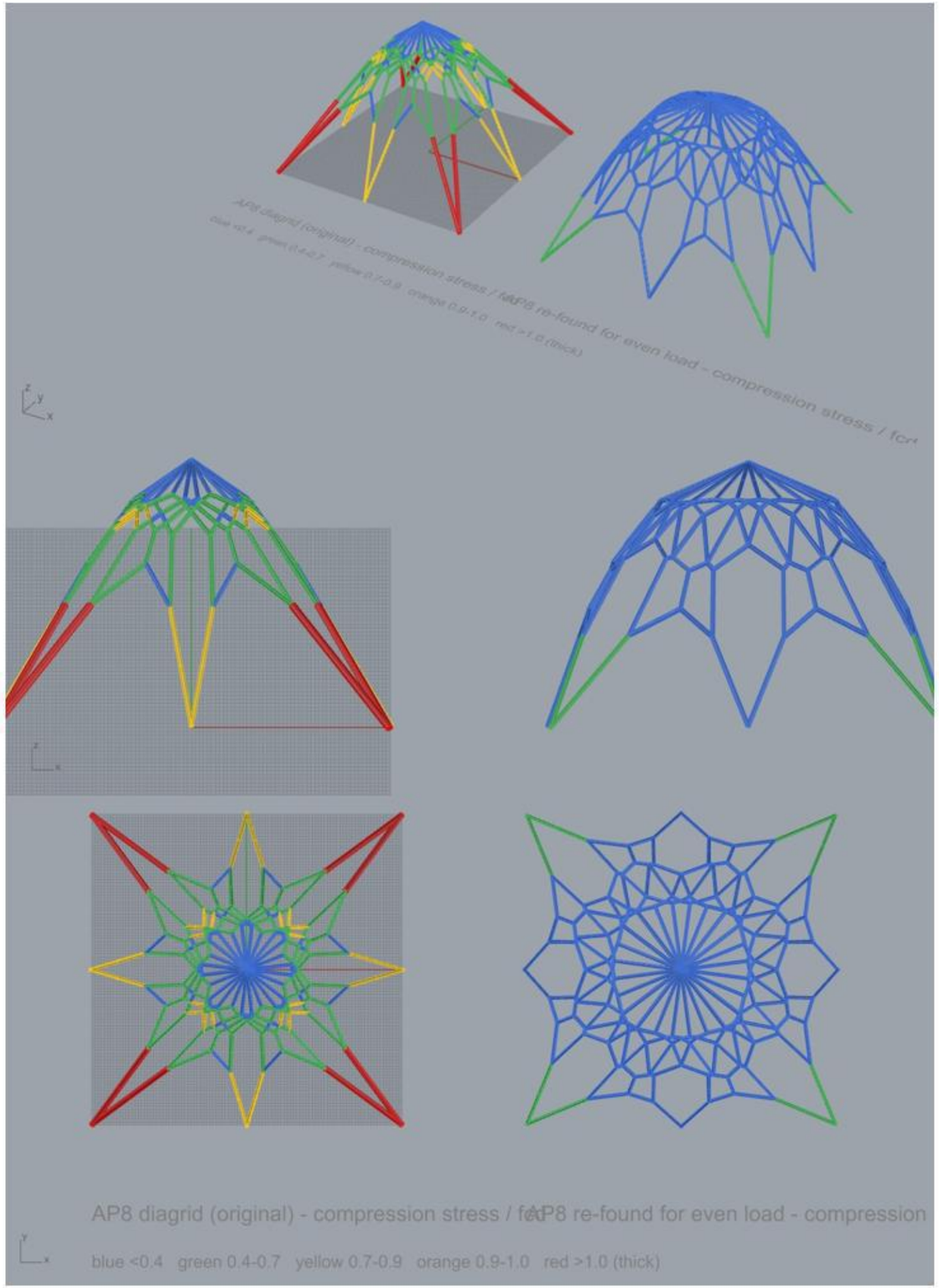
9 16-feet diagrid (5 per side) next to AP8



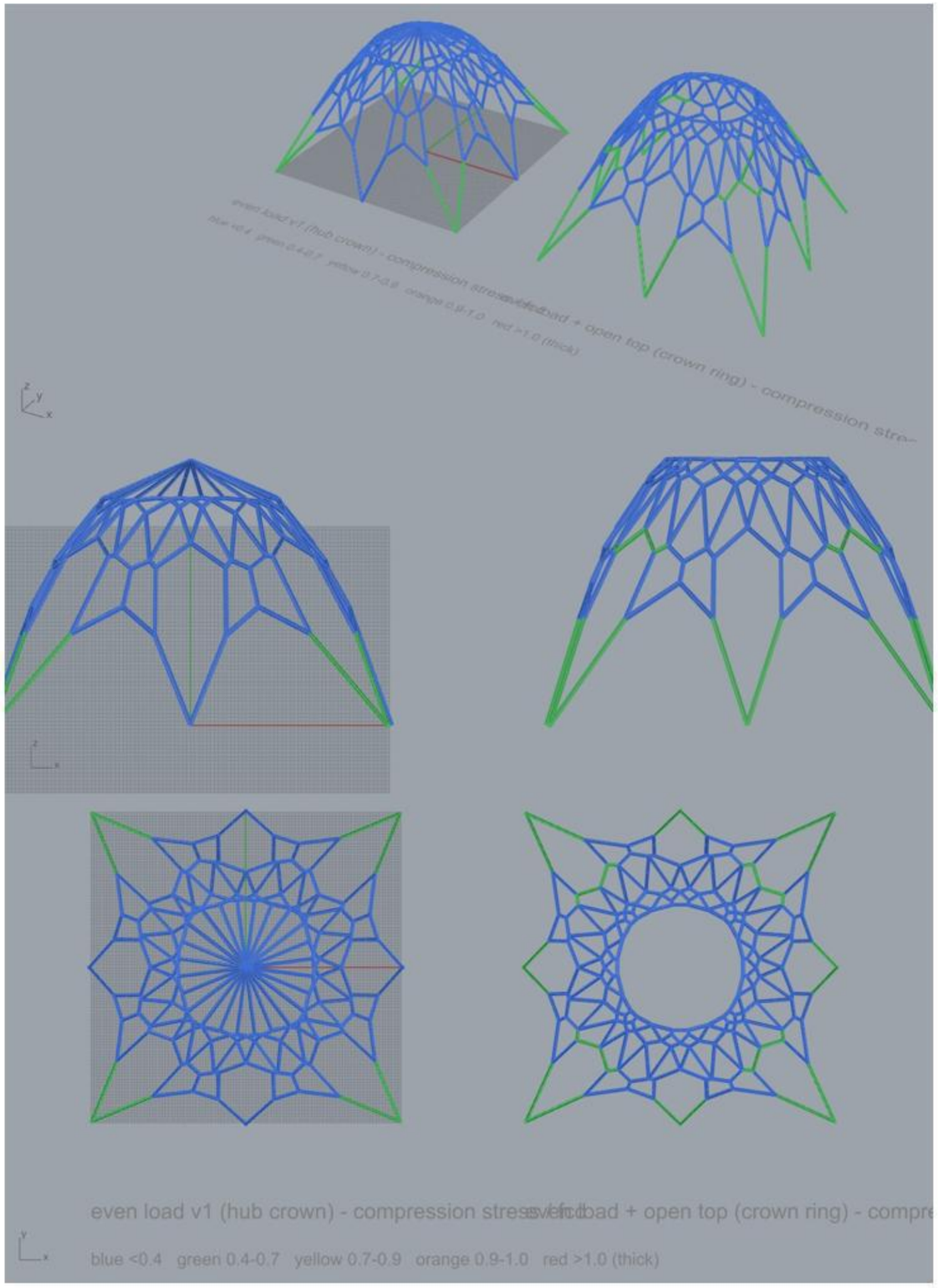
8+9 Karamba check: AP8 diagrid vs 16-feet



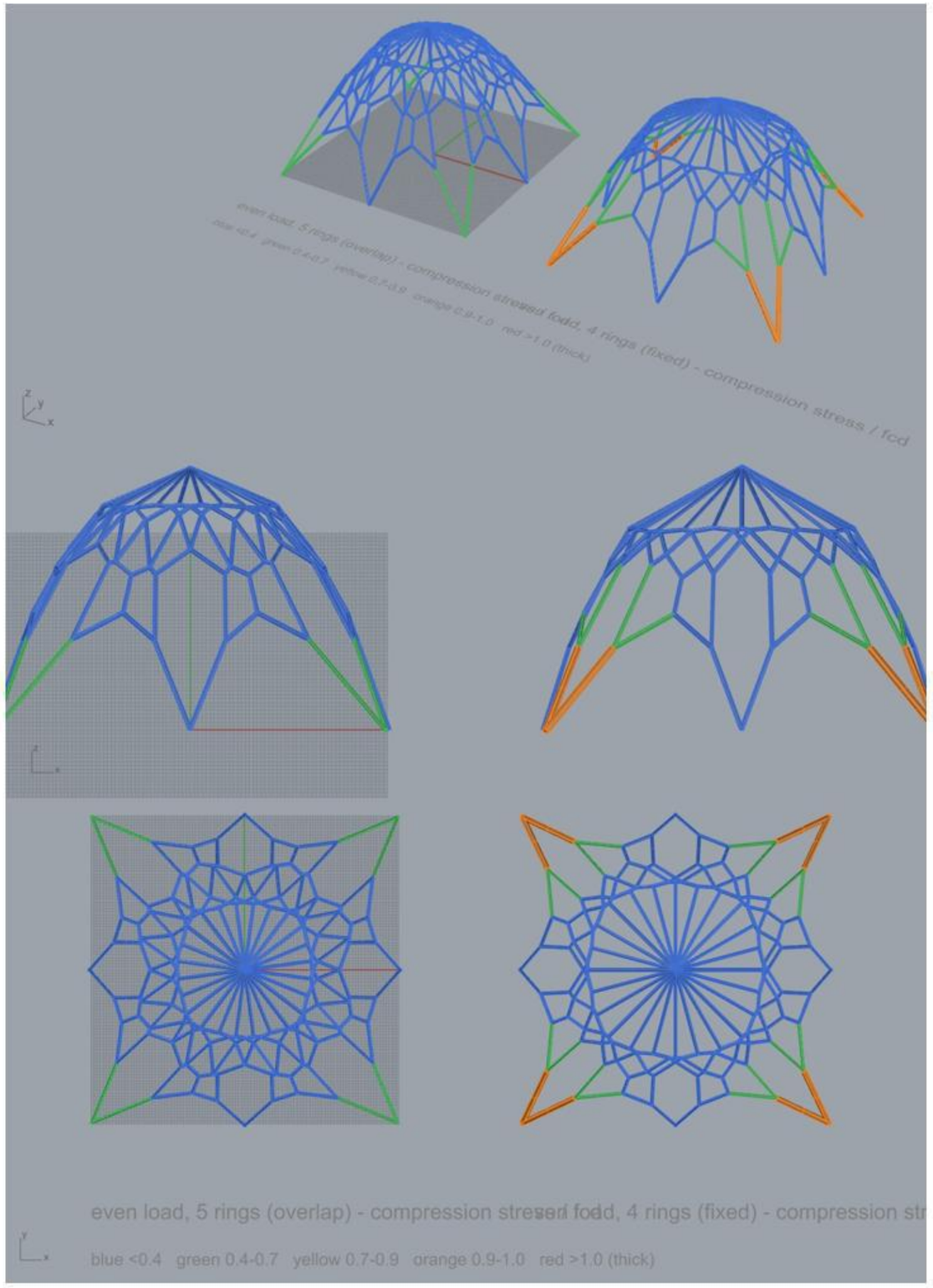
10 AP8 even load (right) vs original (left), Karamba



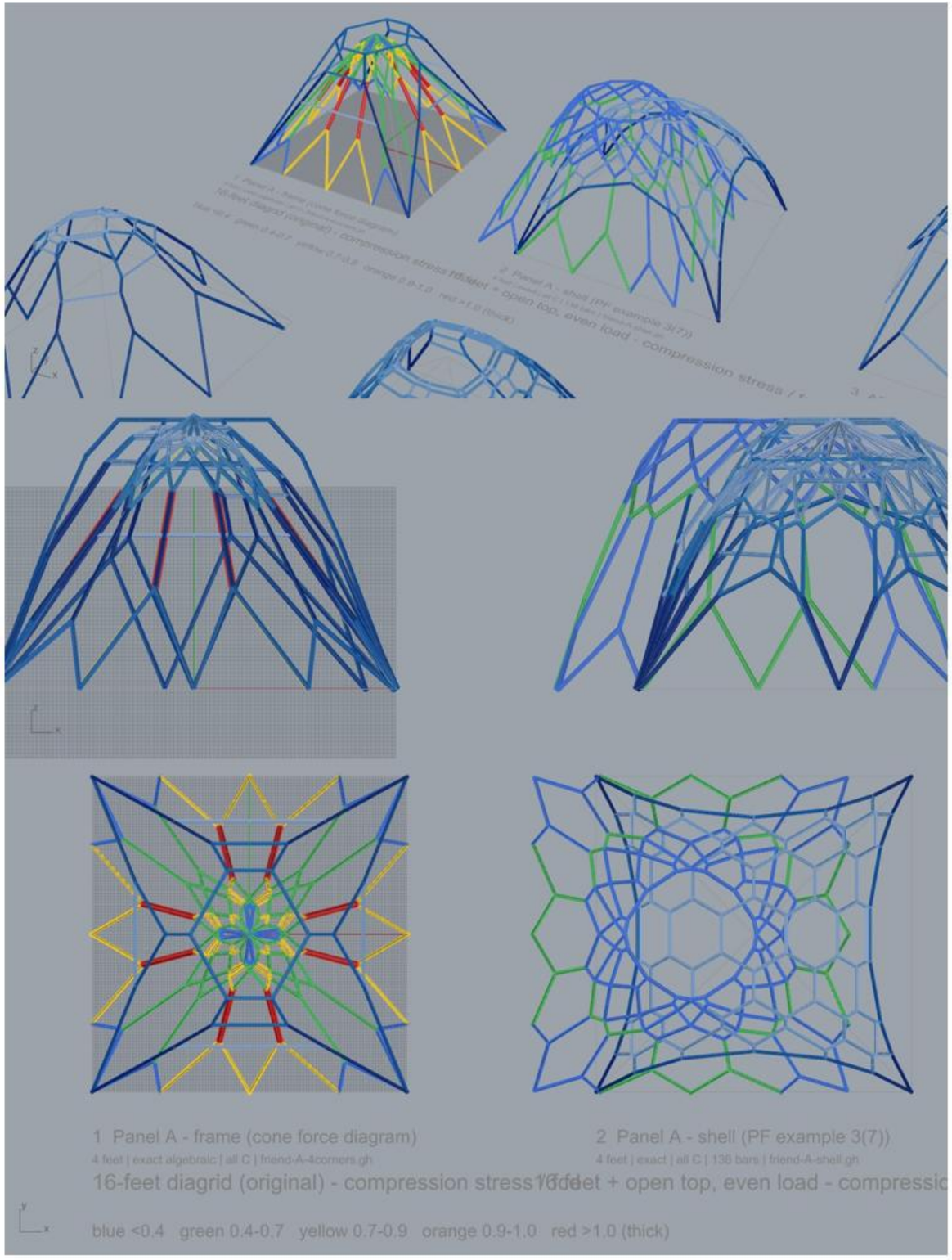
11 OPEN TOP (right) vs even-load hub (left), Karamba



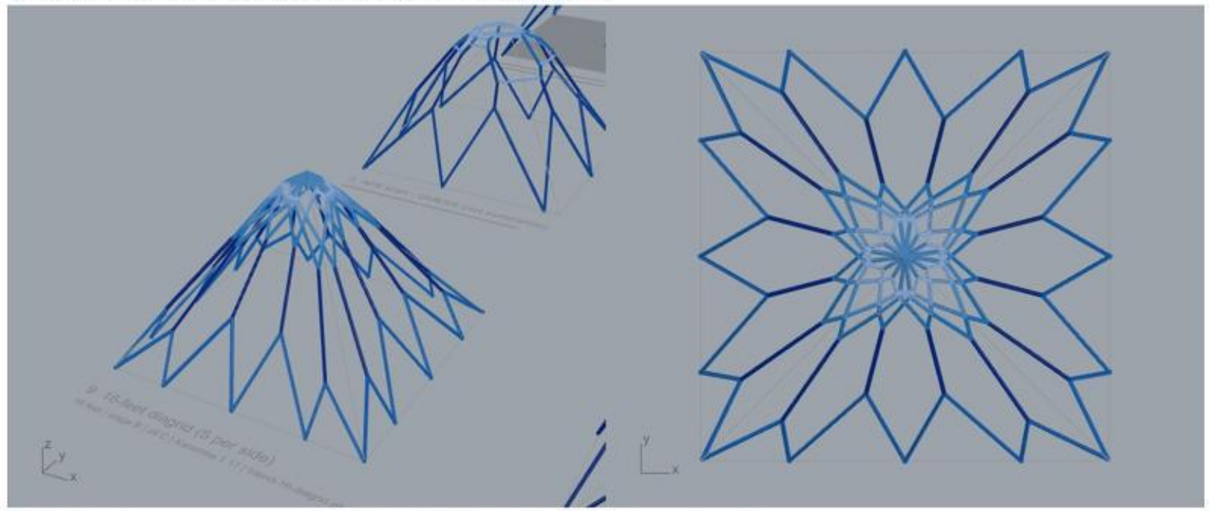
12 4 rings (right) vs 5 rings with overlap (left), Karamba



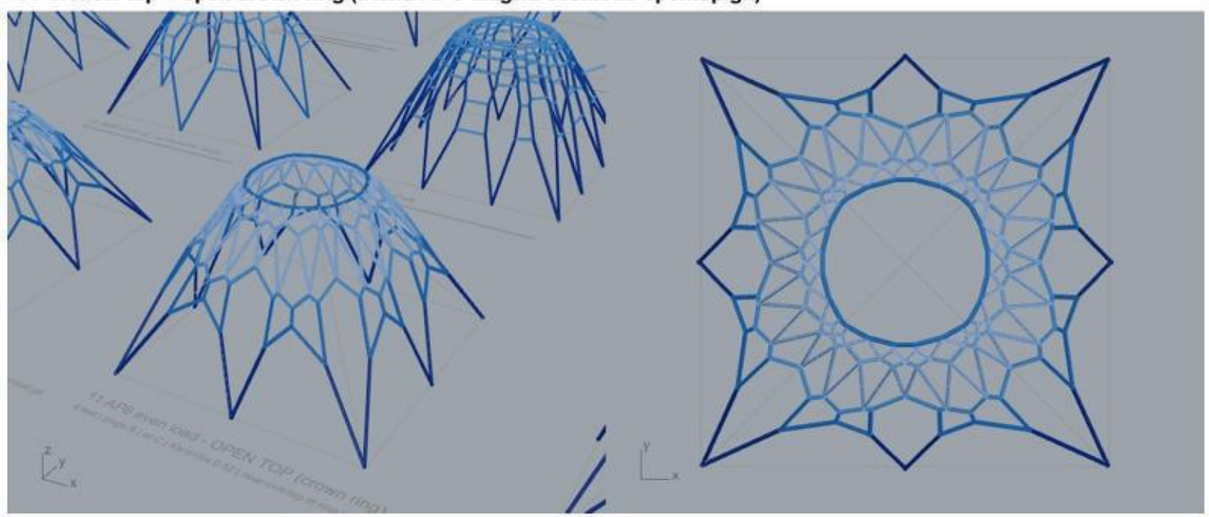
13 16 feet (5 per side) + OPEN TOP, re-found for even load - Karamba vs original 16-feet diagrid (left)



9 16-feet diagrid = 5 supports per side (friend-16-diagrid.gh)

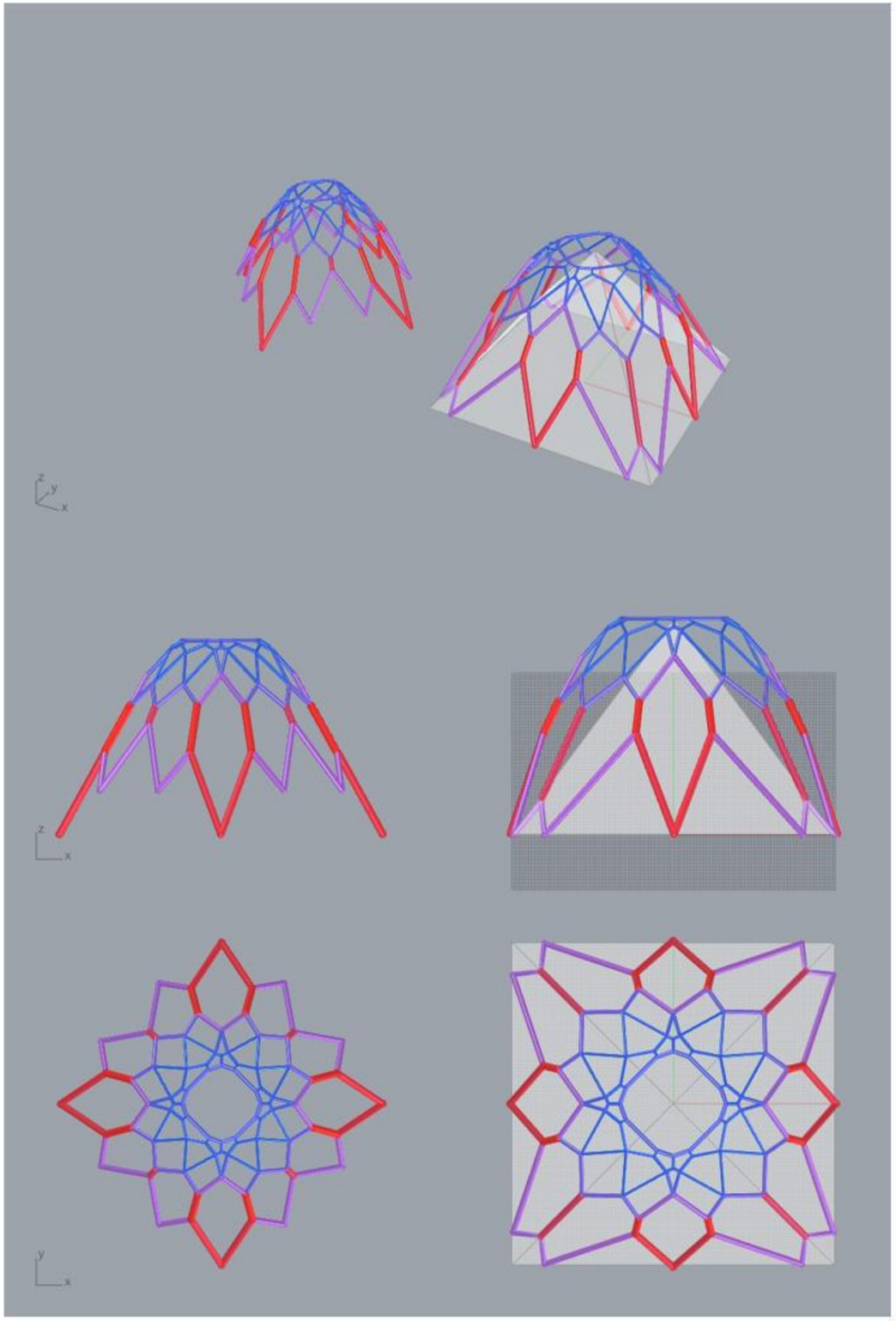


11 Hollow top = open crown ring (friend-AP8-diagrid-evenload-opentop.gh)

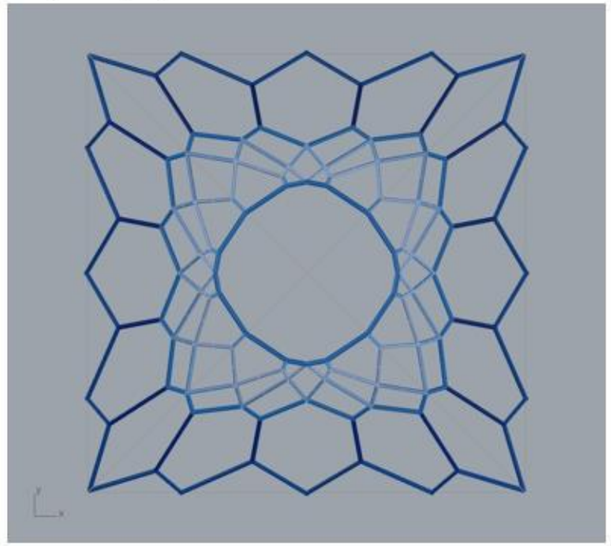


Close-ups of #9 (5 per side) and #11 (hollow top) as they appear in the 3dm

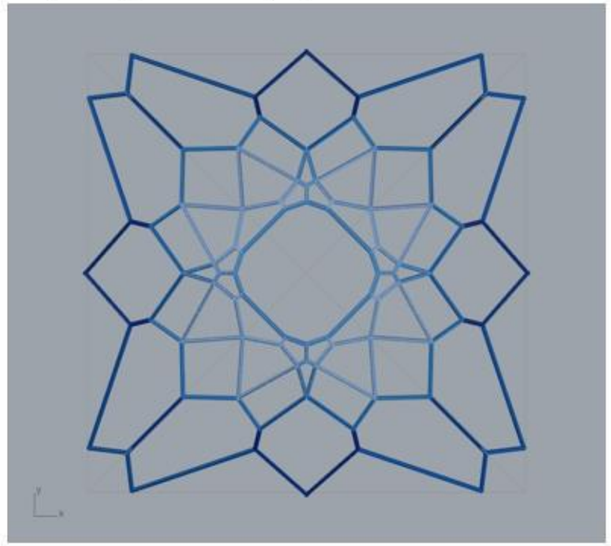
14 12 feet (1 m from each corner + mid-side), corners free, OPEN TOP, re-found for even load



#13 (16 feet) top



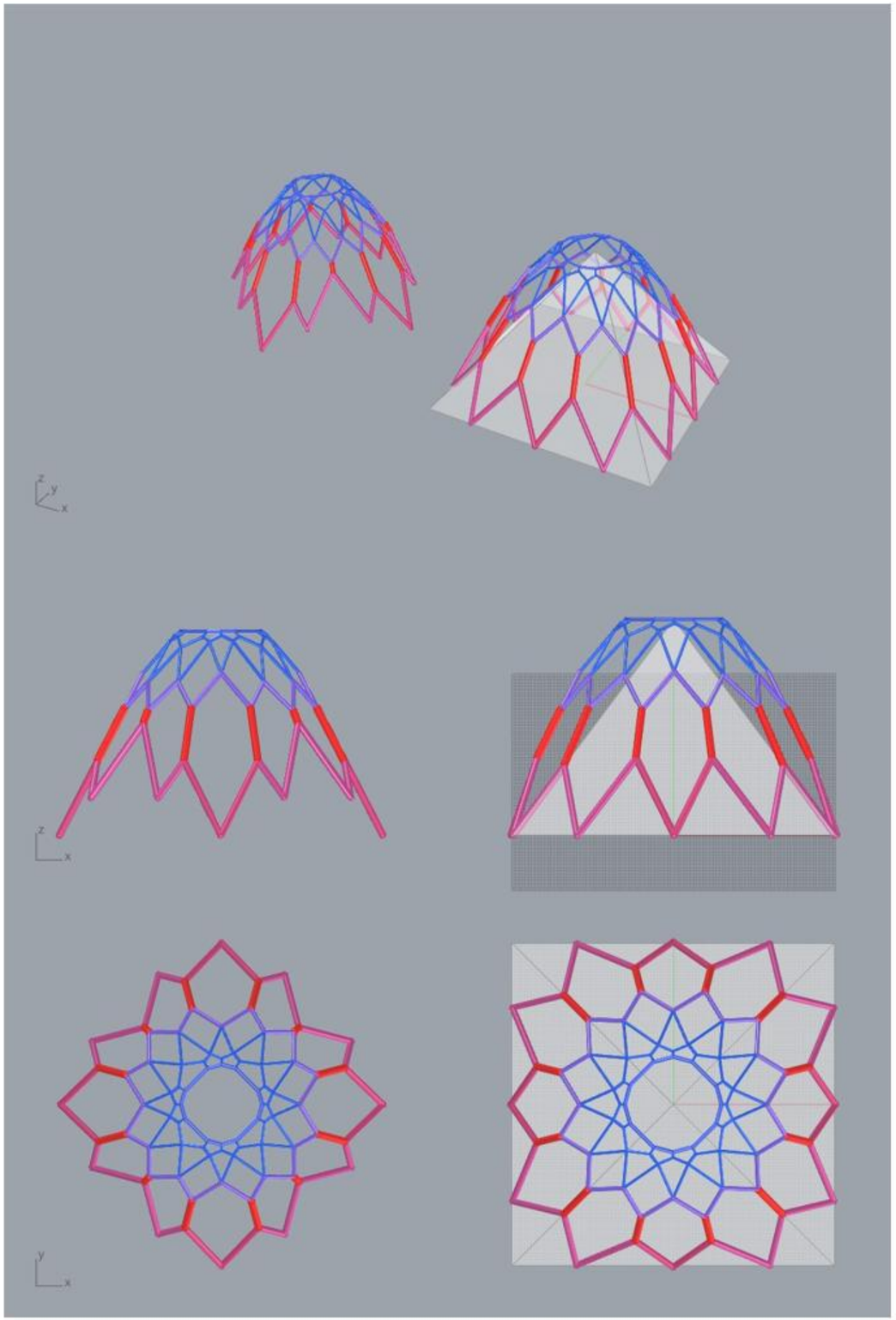
#14 (12 feet, corners free) top



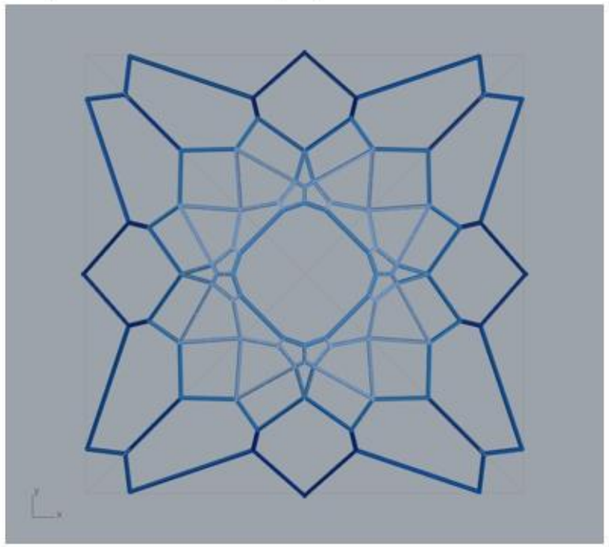
Karamba (150x250 ribs, self-weight + 7 kN/m2)	#13	#14
Max compression / fcd	0.52	0.56
Bending tension / fctm (needs rebar)	2.7	3.1
Max bending	10.9 kNm	10.0 kNm
Axial force range	-54..+2.8	-73..+4.5 kN
Max deflection	5.0 mm	5.6 mm
Buckling factor	38.7	49.3
Rib concrete	21.4 t	18.2 t
Bars / feet	176 / 16	132 / 12
Reactions V/H per foot (kN)	corner 48/32, quarter 90/25, mid 117/25 | #14: near-corner 86/42, mid 165/37	

PolyFrame #14: stage A and B all compression; stage B max deviation 10.1 deg, mean 2.3 deg; feet within 57 mm of targets (mid feet 60 mm outside the wall); crown ring r 1.54-1.64 m at 6.67 m. Corners: net reaches 6.34 m on the diagonal; plan corner is 7.07 m -> 0.73 m not covered. Net rises up to 2.6 m above the 53 deg reference pyramid (it is a dome, not a pyramid). Not verified: 7 kN/m2 load assumed, no wind/seismic, elastic check without rebar.

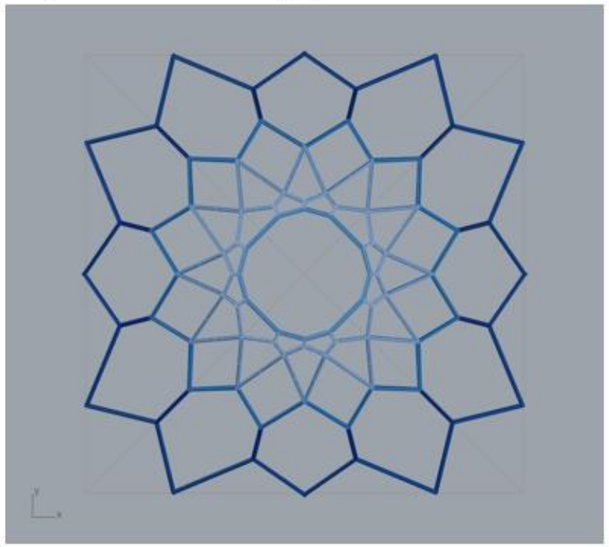
15 12 feet (2 m from each corner + mid-side), corners free, OPEN TOP, re-found for even load



#14 (feet 1 m from corners) top



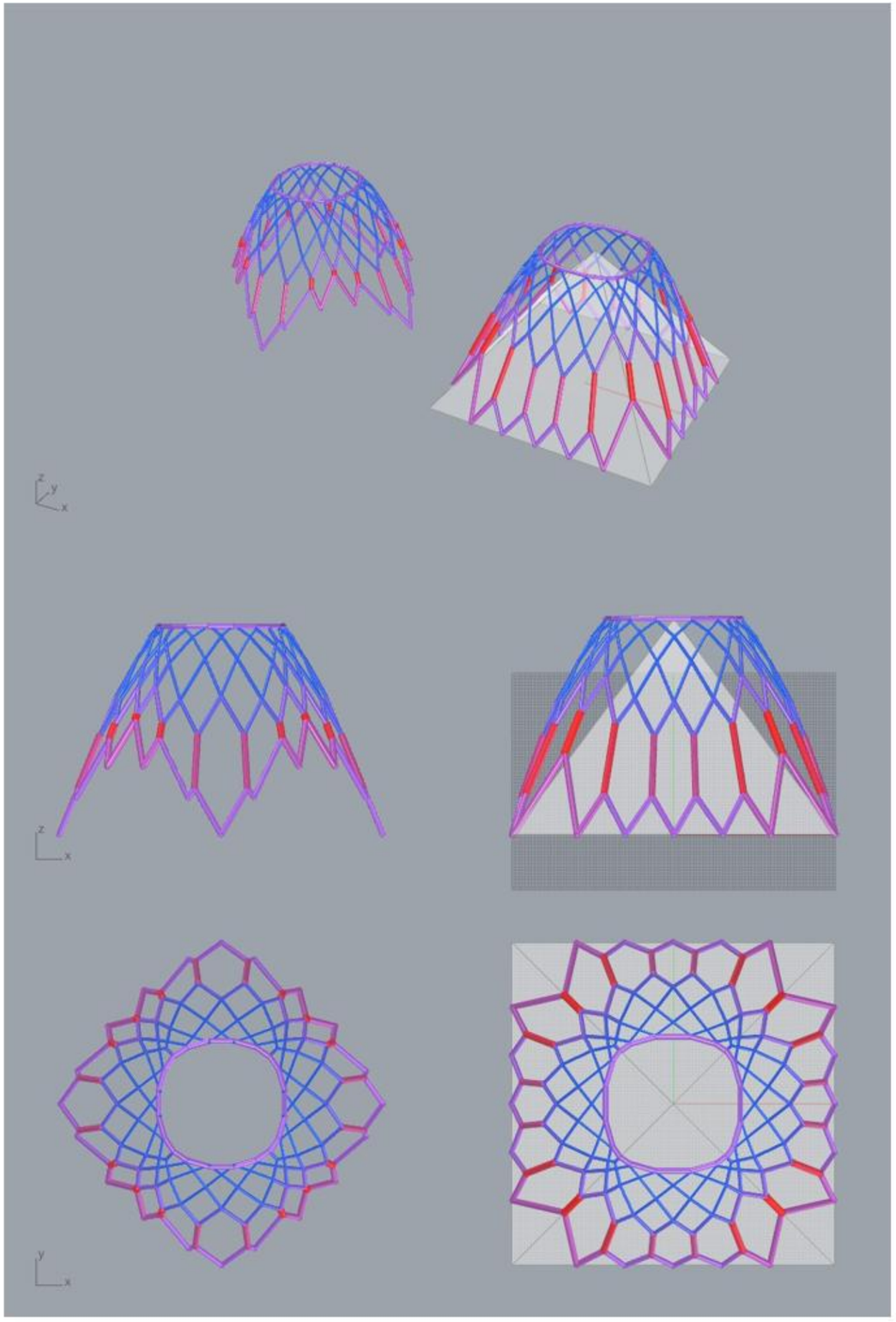
#15 (feet 2 m from corners) top



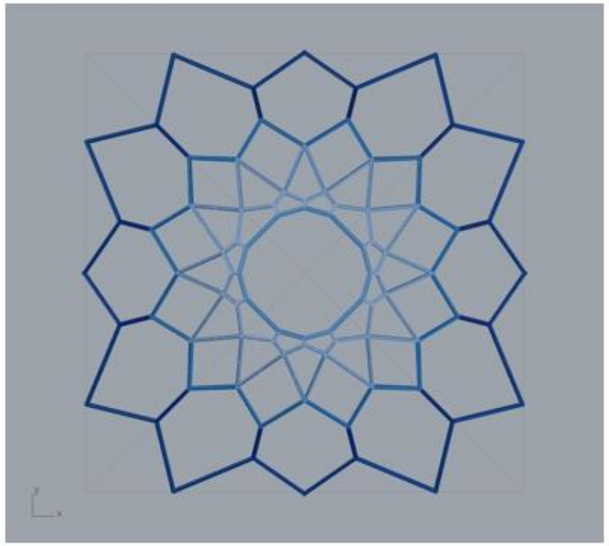
Karamba (150x250, self-wt + 7 kN/m2)	**#14 (1 m)**	**#15 (2 m)**
Max compression / fcd	0.56	0.53
Bending tension / fctm (rebar)	3.1	3.3
Max bending	10.0 kNm	8.3 kNm
Axial force range	-73..+4.5 kN	-62..+2.4 kN
Max deflection	5.6 mm	4.8 mm
Buckling factor	49.3	40.7
Rib concrete	18.2 t	17.7 t
Near-corner foot V / H	86 / 42 kN	107 / 43 kN
Mid-side foot V / H	165 / 37 kN	123 / 28 kN
PolyFrame max / mean deviation	10.1 / 2.3 deg	7.7 / 1.5 deg
Closest node pair	236 mm	191 mm
Net reach on diagonal (corner 7.07 m)	6.34 m	5.64 m
Corner not covered	0.73 m	1.43 m
Height above ref. pyramid	up to 2.6 m	up to 2.3 m

Both: all compression in stage A and B; feet held at exact points. Not verified: 7 kN/m2 assumed, no wind/seismic, elastic check without rebar.

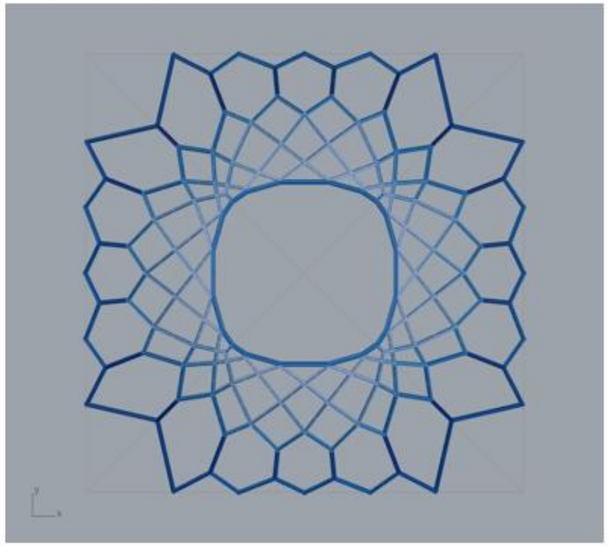
16 20 feet (5 per side, 1.5 m apart, outer 2 m from corners), corners free, **OPEN TOP**, even load



#15 (12 feet) top



#16 (20 feet) top



Karamba (150x250, self-wt + 7 kN/m2)	**#15 (12 feet)**	**#16 (20 feet)**
Total load carried	1,344 kN	1,434 kN
Rib concrete	17.7 t	26.7 t
Max compression / fcd	0.53	0.37
Bending tension / fctm (rebar)	3.3	2.4
Max bending	8.3 kNm	5.9 kNm
Axial force range	-62..+2.4 kN	-61..+9.2 kN
Max deflection	4.8 mm	3.5 mm
Buckling factor	40.7	69.0
Outer foot (2 m from corner) V / H	107 / 43 kN	70 / 24 kN
Quarter foot (1.5 m from mid) V / H	-	64 / 19 kN
Mid-side foot V / H	123 / 28 kN	89 / 21 kN
PolyFrame stage A	all C	has tension (-28 kN)
PolyFrame final max / mean deviation	7.7 / 1.5 deg	11.1 / 2.3 deg
Closest node pair	191 mm	54 mm (16 pairs <200)
Net reach on diagonal (corner 7.07 m)	5.64 m	5.64 m
Height above ref. pyramid	up to 2.3 m	up to 2.8 m

Final forms both all compression; feet held at exact points. Not verified: 7 kN/m2 assumed, no wind/seismic, elastic check without rebar.